Internship Report

on

IMAGE SKETCHER

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY ANANTAPUR, ANANTHAPURAMU

In Partial Fulfillment of the Requirements for the Award of the degree of

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE & ENGINEERING

Submitted By

K. SAI RAJ - 22691A05J1

MADANAPALLE INSTITUTE OF TECHNOLOGY &SCIENCE

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(Affiliated to JNTUA, Ananthapuramu)

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

BONAFIDE CERTIFICATE

This is to certify that the internship work entitled **"Image Sketcher"** is a bonafide work carried out by

K. SAI RAJ - 22691A05J1

Submitted in partial fulfillment of the requirements for the award of degree **Bachelor of Technology** in the stream of **Computer Science & Engineering** in **Madanapalle Institute of Technology & Science, Madanapalle,** affiliated to **Jawaharlal Nehru Technological University Anantapur, Ananthapuramu** during the academic year 2024-2025.

Internship Coordinator	**Head of the Department**
Mr. Ch. Hemanand	Dr.M.Sreedevi,
Assistant Professor,	Professor & Head,
Department of CSE	Department of CSE

Submitted for the University Examination held on:

Examiner - I	**Examiner - II**

ACKNOWLEDGEMENT

We sincerely thank the **MANAGEMENT** of **Madanapalle Institute of Technology & Science** for providing excellent infrastructure and lab facilities that helped me complete this Internship.

We sincerely thank **Dr. C. Yuvaraj, M.E., Ph.D., Principal,** for guiding and providing facilities for completing our Internship at **Madanapalle Institute of Technology& Science,** Madanapalle.

We express our gratitude to **Dr. M. Sreedevi, Ph.D., Professor and Head of the Department, CSE** for her continuous support in making necessary arrangements for the successful completion of the Internship.

We express our sincere thanks to the **Internship Coordinator, Mr. Ch. Hemanand, Assistant Professor, Department of CSE** for his tremendous support for the successful completion of Internship.

We express our deep gratitude to our Internship In-Charge **Ms S.Sowmyadevi, Assistant Professor, Department of CSE,** for his guidance and encouragement that helped us to complete this Internship.

We also wish to place on record my gratefulness to other **Faculty of CSE Department** and our friends and our parents for their help and cooperation during our project work.

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Shri P Abhishek
Human Resources(HR)

Shri K Mukesh Raj
Chief Executive Officer(CEO)
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DECLARATION

We hereby declare that the results embodied in this internship **"Image Sketcher"** by us under the guidance of **Ms.S.Sowmyadevi,** in partial fulfillment of the award of **Bachelor of Technology** in **Computer Science & Engineering** from **Jawaharlal Nehru Technological University Anantapur, Ananthapuramu**.

Date :

Place :Madanapalle

Internship Associate:

K. SAI RAJ

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ABSTRACT

This project demonstrates the conversion of a digital image into a pencil sketch using Python and OpenCV. The process involves several image processing techniques: grayscale conversion, Gaussian blur, edge detection with the Canny algorithm, and binary thresholding to create a sketch-like effect.The transformation begins by loading the image and converting it to grayscale, removing color information while retaining intensity values. This simplifies the image and prepares it for edge detection. A Gaussian blur is then applied to the grayscale image to reduce noise and smooth the image, improving the accuracy of edge detection. The Canny edge detector is used to identify areas of significant intensity change, marking them as edges.Next, binary thresholding is applied to invert the edges, turning them white on a black background, simulating a pencil sketch. This final result is displayed in a other window as sketch output.This project provides an efficient and easy-to-understand method for converting images into artistic sketches using basic computer vision techniques. It highlights how OpenCV's powerful image processing functions can be combined to achieve visually appealing effects, making it suitable for creative applications such as graphic design and digital art.

Keywords : Python,OpenCV,Grayscale,Gaussian blur,Thresholding,Image,Sketch.

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CHAPTER-1

INTRODUCTION

1.1 ABOUT PYTHON

Python is a high-level, interpreted programming language known for its simplicity and readability, making it an ideal choice for both beginners and experienced developers. Created by Guido van Rossum and first released in 1991, Python emphasizes code readability and a straightforward syntax that allows developers to express concepts in fewer lines of code compared to other programming languages.

Key features of Python include:

1.Readable and Simple Syntax: Python's syntax closely resembles human language, making it easy to read, write, and maintain. This is one of the reasons why it is widely used in education and among beginner programmers.

2. Interpreted Language: Python code is executed line-by-line by an interpreter,which makes debugging easier and allows for rapid development and testing.

3. Cross-Platform Compatibility: Python is available on various platforms, including Windows, macOS, and Linux, making it highly portable.

4. Extensive Standard Library: Python comes with a vast standard library that includes modules and packages for handling various tasks, such as file operations, networking, web development, and data processing.

5. Dynamic Typing: Python is dynamically typed, meaning you don't need to declare the type of a variable explicitly. The interpreter determines the type at runtime.

6. Object-Oriented and Functional Programming: Python supports multiple programming paradigms, including object-oriented, imperative, and functionalprogramming styles, giving developers flexibility in their approach.

Significance of Python:

Python is significant due to its simplicity and readability, making it an ideal language for beginners and experienced developers alike. Its concise syntax allows for faster development, reducing the time and effort required to write code. Python supports multiple programming paradigms, including object-oriented and functional programming. It boasts a large standard library and a thriving ecosystem of third- party packages, enabling developers to handle a wide range of tasks efficiently.

Python’s cross-platform compatibility makes it versatile, while its strong community support ensures continuous growth and resource availability. These factors contribute to Python's widespread use in web development, data science, AI, automation, and more.

1.2 IMPORTANCE AND APPLICATIONS OF PYTHON

Python is crucial for image sketching applications due to its simplicity, versatility, and powerful libraries like OpenCV and NumPy. In image processing tasks, such as converting images into pencil sketches, Python's easy-to-understand syntax allows developers to quickly implement and experiment with algorithms. Libraries like **OpenCV** provide pre-built functions for essential tasks such as grayscale conversion, edge detection, and thresholding, which are key steps in generating sketch effects. Additionally, Python’s strong support for scientific computing and data manipulation makes it ideal for handling complex image processing workflows efficiently. This makes Python an excellent tool for both simple and advanced image processing projects, including artistic transformations like sketching.

Python, with its powerful libraries and ease of use, is widely applied in creating image sketch effects and other image manipulation tasks. Some of the key applications are:

- Artistic Image Transformations
- Photo Editing Software
- Web-Based Image Processing
- Machine Learning for Style Transfer
- Automated Design Tools
- Photo Restoration and Enhancement
- Mobile Applications

1.3 LANGUAGE USED

PYTHON

Python language is used to implement the image sketcher

It uses the **OpenCV library** (specifically cv2), which is a popular computer vision library, to perform image processing tasks like converting an image to a sketch. The key operations used in the code are:

- **Image loading** with cv2.imread()
- **Grayscale conversion** with cv2.cvtColor()
- **Blurring** with cv2.GaussianBlur()
- **Edge detection** using **Canny Edge Detection** with cv2.Canny()
- **Thresholding** using cv2.threshold()
- **Displaying the image** with cv2.imshow()

1.4 NEED FOR THE MODEL

To run the Python code for converting an image to a pencil sketch using OpenCV, you will need the following:

1. Software Requirements

- **Python**: Python version 3.x is recommended. You can download Python from the official website: Python Downloads.
- **OpenCV Library**: The code uses OpenCV, which is an open-source computer vision library. You can install OpenCV using pip (Python's package installer).

  You can install OpenCV by running the following command in your terminal or command prompt:

  pip install opencv-python

2. Hardware Requirements

- **Computer with an image processing-capable CPU**: This code does not require heavy resources, but for faster processing, a modern CPU is preferable.
- **Memory (RAM)**: A system with at least 2GB of RAM should be enough for basic image processing tasks like the one in your code. More RAM is required for processing larger images or performing real-time processing.

3. Image Files

- **Input Image**: The script takes an input image (which you need to load) to process and convert into a pencil sketch. This image should be in a supported format such as PNG, JPG, or JPEG.

4. **Python Libraries**

Besides OpenCV, you may also need **NumPy**, as OpenCV functions work in conjunction with NumPy arrays.

You can install NumPy using:

bash

Copy code

```
pip install numpy
```

5. Basic Knowledge of Python

- Understanding basic Python syntax, functions, and how to work with libraries will help in modifying and running the code.
- **File Handling**: You'll need to know how to load image files into Python using cv2.imread().

6. IDE (Integrated Development Environment)

- You can run the code in any Python IDE or editor such as:
  - **VSCode**: A powerful code editor.
  - **PyCharm**: A popular Python-specific IDE.
  - **Jupyter Notebooks**: Ideal for data science tasks and testing small code snippets.

Alternatively, you can also run the script directly from a terminal if you prefer.

7. Operating System Compatibility

- The code is compatible with all major operating systems (Windows, macOS, and Linux). Ensure that Python and OpenCV are installed correctly for your specific operating system

CHAPTER-2

TOOLS AND TECHNIQUES

2.1 Platform used:

1. Local Python Environment

- **Python IDLE:** You can run the code locally on your computer using the IDLE (Integrated Development and Learning Environment) that comes bundled withPython.
- **Jupyter Notebook**: This is a great environment for writing and testing Python code interactively. It allows for easy testing of code and visualization.
- **VS Code / PyCharm / Sublime Text**: These are full-fledged IDEs that support Python development and come with features like debugging, codecompletion, and more.

2. Online Platforms

If you don't want to set up anything locally, there are online platforms where you canwrite and run Python code:

- **Replit:** An online platform that allows you to write and run Python code in aweb browser. You can also share your game with others.

  Website: replit.com

- **Google Colab**: A cloud-based Python development environment by Google that is great for running Python code and experimenting with machine learningas well.

  Website: colab.research.google.com

- **Trinket:** Another platform to write and run Python code directly in thebrowser.

  Website: trinket.io

- **Programiz**: An easy-to-use online compiler for running Python programs, including basic games.

  Website: programiz.com

2.2 Hardware Requirements:

1.Processor (CPU)

- **Minimum**: Any modern CPU with at least 2 cores (e.g., Intel Core i3 or AMD Ryzen 3).
- **Recommended**: Multi-core processors (e.g., Intel Core i5/i7 or AMD Ryzen 5/7) for faster image processing, especially if working with high-resolution images or processing multiple images at once.

2.Memory (RAM):

- **Minimum**: 4 GB of RAM.
- **Recommended**: 8 GB or more. This allows for smoother processing of larger images or running multiple image processing tasks simultaneously.

3.Graphics Processing Unit (GPU):

- **Minimum**: A GPU is not strictly required for this project, as the operations used (grayscale conversion, edge detection, thresholding) are not computationally intensive.
- **Recommended**: A GPU is useful if you plan to process very high-resolution images or use more complex image processing techniques, such as neural networks or deep learning-based style transfer. A basic GPU like NVIDIA's GeForce GTX or AMD’s Radeon series would suffice in such cases, but again, it's not necessary for this simple sketching project.

4.Storage:

- **Minimum**: 1 GB of free disk space to store images and the necessary libraries.

- **Recommended**: 5 GB or more if you're working with large image datasets or need space for storing processed images.

5.Operating System:

- **Windows**: Windows 10 or later (64-bit recommended).
- **Linux**: Any modern distribution, such as Ubuntu 20.04 or later.
- **macOS**: macOS 10.15 (Catalina) or later.

2.3 Software Requirements:

1. **Python**:

- Python 3.x (preferably 3.6 or later) installed.

2. **Libraries**:

- **OpenCV**: For image manipulation and processing (pip install opencv-python).
- **NumPy**: For numerical operations and matrix manipulation (pip install numpy).
- **Matplotlib** (optional, for displaying images in Jupyter/Colab): pip install matplotlib.

CHAPTER-3

PROJECT WORK

3.1 Project Overview

Image sketcher demonstrates the use of Python and OpenCV to transform digital images into pencil sketch effects. The main goal is to take an image as input, apply various image processing techniques, and output an image that mimics a hand-drawn pencil sketch. The process involves converting the image to grayscale, blurring it to smooth out details, detecting edges, and finally inverting those edges to create a sketch-like effect.

Objective:

The objective of this project is to provide a Python-based tool that can automatically generate pencil sketch effects from images. By leveraging the power of OpenCV, the project showcases a variety of image processing techniques that allow users to convert images into stylized artwork. This effect is useful in various creative applications, such as enhancing photos, creating artistic content, and designing promotional materials.

Key Features:

- **Image Conversion**: Convert input images into grayscale.
- **Edge Detection**: Detect edges using the Canny edge detection algorithm.
- **Sketch Effect**: Apply thresholding to create a pencil sketch effect.
- **Easy-to-Use**: Simple Python script that takes an image file and outputs a sketch.
- **Customizable**: Potential to adjust processing parameters such as blur intensity and edge detection thresholds for different artistic results.
- Technologies Used:
- **Python**: A versatile programming language that allows easy integration of libraries and tools for image processing.

- **OpenCV**: A powerful open-source library used for real-time computer vision tasks, such as reading, processing, and transforming images.
- **NumPy**: A library for numerical operations, used by OpenCV for handling image data.
- **Matplotlib (Optional)**: For displaying the resulting sketch in Jupyter or Colab environments.

3.2 Algorithm

1. **Input Image:**
    - Take an input image (img), which is in color (BGR format).
2. **Convert Image to Grayscale**:
    - Convert the input color image to a grayscale image.
    - This simplifies the image to intensity levels (black, white, and gray), making it easier to detect edges.
3. **Apply Gaussian Blur**:
    - Apply a Gaussian blur to the grayscale image to reduce noise and smooth the image. This helps in detecting edges more effectively.
    - The cv2.GaussianBlur() function is used with a kernel size of (5,5) or larger, depending on the desired effect.
4. **Edge Detection using Canny Algorithm**:
    - Detect edges in the blurred grayscale image using the Canny edge detection algorithm.
    - The Canny edge detection algorithm detects areas of high contrast and identifies the boundaries of objects.

5. **Invert the Edges**:
    - Invert the edges image to create a white sketch effect on a black background.
    - This is done because the Canny edge detector produces white edges on a black background, which we want to reverse for the sketch effect.
6. **Return the Final Sketch Image**:
    - The resulting image is the inverted edge-detected image, which appears as a pencil sketch on a white background.
    - Display the final image using cv2.imshow() or Matplotlib for visualization.

Pseudocode:

1. Load the input image 'img' using OpenCV (cv2.imread).

2. Convert the image to grayscale (cv2.cvtColor).

3. Apply Gaussian Blur to the grayscale image (cv2.GaussianBlur).

4. Detect edges using Canny edge detection (cv2.Canny).

5. Invert the edges (cv2.threshold).

6. Display the resulting pencil sketch using cv2.imshow or Matplotlib.

CHAPTER-4

Code and output

screenshots

Source code:

```
# To install opencv

!pip install opencv-python

# To import modules

import cv2

import numpy as np

# Function to convert an image to a sketch effect

def sketch(img):

    # Convert the image to grayscale

    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Apply Gaussian blur to the grayscale image

    gray_blur = cv2.GaussianBlur(gray, (5, 5), 0)

    # Use the Canny edge detector

    canny_edges = cv2.Canny(gray_blur, 10, 70)

    # Invert the edges to get a white sketch on a black background

    r, mask = cv2.threshold(canny_edges, 70, 255, cv2.THRESH_BINARY_INV)

    return mask
```

```python
# Load an image from a file
image_path = ".png"  # Enter image path
img = cv2.imread(image_path)
# Check if the image was loaded successfully
if img is None:
    print(f"Error: Could not load image from {image_path}")
else:
    # Apply the sketch filter
    sketched_img = sketch(img)
    # Display the sketched image using cv2.imshow
    cv2.imshow("Pencil Sketch", sketched_img)
    # Wait until a key is pressed, then close the displayed window
    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

INPUT 1:

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#startupindia

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Slash Mark IT Startup

Intern ID : SMI70162

OUTPUT 1:


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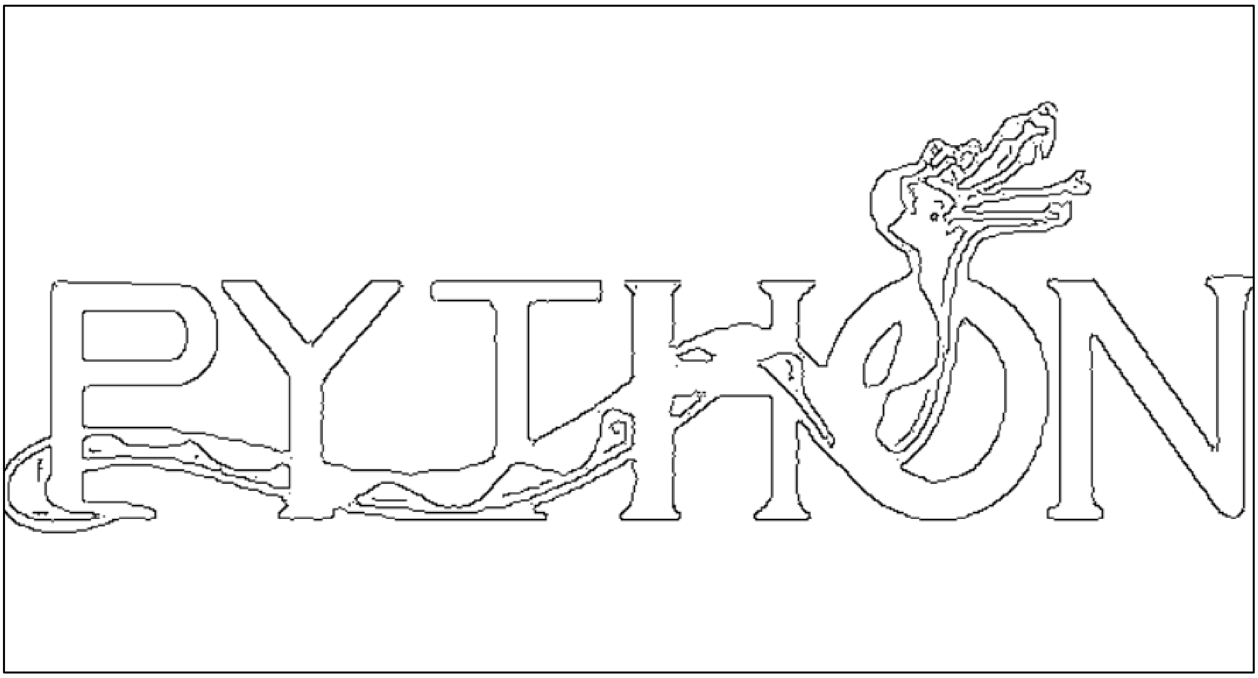
Slash Mark IT Startup
Intern ID : SM170162

INPUT 2:

OUTPUT 2:



CHAPTER-5

Conclusion

Conclusion:

Image Sketcher demonstrates the effectiveness of Python and OpenCV in transforming a simple image into a pencil sketch using fundamental image processing techniques. The core of the project involves four primary steps: converting the image to grayscale, applying a Gaussian blur to reduce noise, detecting edges using the Canny edge detector, and finally, inverting the edges to create a pencil sketch effect. These processes are executed efficiently using Python's extensive libraries, making it an ideal choice for both beginners and experienced developers looking to perform basic to intermediate image manipulation tasks.

The simplicity of Python, combined with the power of OpenCV, allows for easy experimentation with different parameters, such as the intensity of the blur or edge detection thresholds, enabling users to customize the output for various artistic results. This ability to adjust the settings makes the project highly adaptable, allowing users to fine-tune the effect to match specific requirements or creative preferences.

From a broader perspective, the project showcases how Python can be used not only for basic image processing tasks but also for more complex applications in fields like computer vision, digital art, and photography. The pencil sketch effect generated here could be further extended into more advanced creative tools, such as incorporating machine learning models for style transfer or applying other artistic filters to images.

Moreover, this project highlights the importance of learning and applying core image processing techniques, which are foundational in numerous modern applications like photo editing software, automated design tools, and even artificial intelligence.

By utilizing Python and OpenCV, developers and artists alike can unlock a wide range of possibilities for automating and enhancing image transformations, making it a valuable skill in today’s digital landscape.

Overall, this project is a successful example of leveraging Python’s capabilities to create visually compelling and creative image effects with minimal computational requirements.

REFERENCE:

- **"Learning OpenCV 4" by Adrian Kaehler and Gary Bradski**: This book is one of the most comprehensive resources for learning OpenCV. It covers all the core concepts, including edge detection and various image transformations.
- **"Python Computer Vision with OpenCV" by Joseph Howse**: This book is an excellent resource for learning how to use OpenCV with Python to create various computer vision applications, including image processing effects like sketches.

- Python Official Documentation: A valuable resource for understanding Python's built-in functions and libraries.
  - Python Documentation
  - Specifically for the random module: random module documentation

- Python.org - Python Tutorial: A great starting point for understanding Python programming basics.
  - Python Tutorial
- Real Python: Offers excellent tutorials on various Python topics, including games and projects.
  - Real Python
- Coursera - Python for Everybody: A course that covers Python fundamentals and can be helpful for understanding concepts such as loops, functions, and modules.
  - Python for Everybody - Coursera
- Udemy - Python Programming for Beginners: A beginner-friendly course that could serve as a reference for game development in Python.
  - Python for Beginners - Udemy